Soft Computing Lab (AIMLPPC108)

Basic Assignment 4

a) Practice different Defuzzification methods
b) Practice different Fuzzy Inference Systems

Problem Sheet 3

1. Loan Approval Decision System

A bank evaluates loan applications based on Applicant Income, Credit Score, and Employment Stability. These inputs are fuzzified into linguistic terms such as Low, Medium, and High. A fuzzy inference system is designed to compute the Loan Approval Strength. Students must implement the system using Mamdani FIS with fuzzy output sets and Sugeno FIS with weighted linear outputs. The final approval decision is compared for both approaches.

2. Temperature Control in Smart Buildings

A smart HVAC system adjusts cooling based on Room Temperature, Humidity Level, and Occupancy Density. Fuzzy rules are constructed to determine the Cooling Power Level. Using Mamdani FIS, the output is a fuzzy set requiring defuzzification. Using Sugeno FIS, the output is a crisp value obtained through weighted averaging. Students analyze response smoothness and computational cost.

3. Medical Risk Assessment System

Doctors assess patient health risk using Blood Pressure, Blood Sugar Level, and Body Mass Index (BMI). These inputs are modeled using appropriate membership functions. The fuzzy system estimates a Health Risk Index. Mamdani FIS provides interpretable linguistic outputs, while Sugeno FIS produces numerical risk scores. The two approaches are compared for clinical decision support.

4. Student Academic Performance Evaluation

A university evaluates student performance using Attendance Percentage, Internal Assessment Score, and Assignment Quality. A fuzzy inference system predicts the Final Performance Level. Mamdani FIS outputs linguistic grades, whereas Sugeno FIS outputs a numerical performance index. Students compare interpretability and ease of ranking students.

5. Traffic Congestion Prediction

Urban traffic congestion is estimated using Vehicle Density, Average Speed, and Road Width. These inputs are fuzzified to infer a Congestion Level. Mamdani FIS uses fuzzy congestion categories, while Sugeno FIS computes a congestion score. The systems are compared in terms of real-time suitability.

6. Crop Irrigation Decision System

An agricultural system decides irrigation amount based on Soil Moisture, Temperature, and Crop Growth Stage. Fuzzy rules determine the Water Requirement. Mamdani FIS provides linguistic irrigation levels, whereas Sugeno FIS produces exact irrigation quantities. Students compare accuracy and practical usability.

7. Employee Performance Appraisal

An organization evaluates employees using Work Quality, Timeliness, and Team Contribution. A fuzzy inference system computes an Overall Performance Score. Mamdani FIS outputs fuzzy performance categories, while Sugeno FIS gives a numerical appraisal score. The outputs are compared for HR decision-making.

8. Air Quality Index Estimation

Air quality is assessed using PM2.5 Concentration, NO_2 Level, and Humidity. A fuzzy system estimates the Air Quality Index (AQI). Mamdani FIS produces linguistic AQI categories, while Sugeno FIS outputs a crisp AQI value. Students study sensitivity and numerical stability.

9. Vehicle Speed Control System

An intelligent vehicle adjusts speed using Road Condition, Traffic Density, and Visibility Level. The fuzzy system determines the Recommended Speed Adjustment. Mamdani FIS emphasizes rule interpretability, while Sugeno FIS supports faster real-time computation. The performance of both systems is compared.

10. Customer Satisfaction Prediction

A service provider evaluates customer satisfaction based on Service Quality, Response Time, and Cost Effectiveness. A fuzzy inference system estimates Overall Satisfaction Level. Mamdani FIS yields linguistic satisfaction grades, while Sugeno FIS generates a numerical satisfaction index. Students compare decision clarity and scalability.

---XXXX---